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ABSTRACT

The implementation of a new program for the variational calculation of rovibrational state energies and infrared intensities is presented. The program relies on vibrational self-consistent field and vibrational configuration interaction theory and is based on the Watson Hamiltonian. All needed prerequisites, i.e., multidimensional potential energy and dipole moment surfaces, comprehensive symmetry information, the determination of vibrational wave functions, and an efficient calculation of partition functions, are computed in a fully automated manner, which allows us to calculate rovibrational spectra in a black-box type fashion. Moreover, the use of a molecule specific rotational basis leads to reliable rovibrational line lists. Benchmark calculations are provided for thioformaldehyde (H_2CS), which shows strong Coriolis coupling effects and a complex rovibrational spectrum. The underlying multidimensional potential energy surface has been calculated at the level of explicitly correlated coupled-cluster theory.

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I. INTRODUCTION

The accurate calculation of vibrational spectra beyond the harmonic approximation became more and more popular within the last decades.[1–5] The reason for this is the ever-increasing compute power and the further development of highly efficient and accurate electronic structure methods for calculating multidimensional potential energy surfaces. This motivated intensified activities concerning the development of refined quantum chemical methods for the precise simulation of vibrational spectra. These cover a wide range of approaches, for example, second and fourth order vibrational perturbation theory,[6–8] vibrational coupled-cluster[9] and linear response methods,[10,11] different variants of single and multi-reference vibrational configuration interaction (VCI) approaches,[12–15] and so forth.[16] Most of these developments focus on the determination of purely vibrational transitions ($J = 0$) for non-rotating molecules. Contrarily, the most accurate experimental reference data are obtained from gas phase measurements, which are well known for strong rotational side bands. Moreover, astrochemical studies primarily rely on rovibrational line lists, and thus, there is a high need for methods capable of calculating these desired transitions and their intensities with high accuracy; see, for example, Ref. 17 and the references therein. Of course, program packages are around offering exactly this. For example, TROVE program of Yurchenko and co-workers,[18] the MULTIMODE code of Bowman and Carter,[19–21] TENSOR and MIRS programs by Rey *et al.*[22] and Nikitin *et al.*,[23] or the GENIUSH package of Matyus *et al.*[24] allows for rovibrational calculations, and surely, this short list of pioneering codes is incomplete.[25] To the best of our knowledge, none of these highly advanced packages offers the possibility to compute a rovibrational spectrum in a black-box style by just providing geometrical information about the molecule together with a very limited number of keywords for controlling the calculations. Either one needs to provide a potential energy surface in a predefined format or very specific information about the rovibrational structure calculations is needed, requiring the user to have substantial insight into the

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computational details. The philosophy behind the approach presented here is different: We intend to provide a largely automated approach, allowing for the robust and self-contained calculation of rovibrational spectra for arbitrary molecules from the very beginning. This, of course, requests a high reliability of the working algorithms, which, in the worst case, may be much slower than highly specialized approaches, often working only for specific molecules or in certain situations.

The approach presented here relies on a series of well established methodologies and approaches,[20] but also some specific aspects, the latter to be outlined in detail below. For example, an n-mode expansion[26] of the multi-dimensional potential energy surface (PES) and dipole moment surface (DMS) will be used throughout. This representation allows for a highly efficient and fully automated determination of the surfaces. Symmetry information, which is integral to rovibrational spectra, can be accounted for within the surface generation for arbitrary molecular point groups. Moreover, the Watson Hamiltonian[27] will be employed, which performs well for a multitude of molecules. Vibrational wave functions will be determined from configuration-selective VCI calculations relying on a real basis of Hartree products obtained from vibrational self-consistent field (VSCF) calculations; for details, see Refs. 14 and 28. The rovibrational features are a recent development and will be outlined below. The basic principles may be well-established, but are presented for completeness, while other aspects are specific to our approach. All parts sketched here have been implemented in the MOLPRO package of *ab initio* programs.[29] Benchmark calculations will be provided for thioformaldehyde, which allows for a direct comparison with experimental and computational results of other research groups. This molecule is known to show an accidental near-degeneracy of two vibrational modes and strong Coriolis[30] coupling, leading to a complex spectrum well suited for a benchmark.

II. METHODS

Rovibrational infrared intensities I_{if} are computed as follows:

$$I = \frac{2\pi^2}{3}\frac{N_{\mathrm{A}}}{\varepsilon_0 h^2 c^2}\frac{e^{-E''/kT}(1-e^{-(E'-E'')/kT})}{Q(T)}(E'-E'')R, \tag{1}$$

with the first two prefactors containing only (natural) constants, a temperature-dependent factor with the thermal occupancies of the initial (energy E'') and final (energy E') states and the partition function $Q(T)$, the frequency $(E'-E'')$ of the transition, and transition moments R. In the following, the computation of each of these components will be described in some detail.

A. Rovibrational energies

As mentioned above, throughout this work, we employ the full Watson operator,[27]

$$H = \frac{1}{2}\sum_{\alpha\beta}(J_\alpha - \pi_\alpha)\mu_{\alpha\beta}(J_\beta - \pi_\beta) - \frac{1}{2}\sum_i \frac{\partial^2}{\partial q_i^2} - \frac{1}{8}\sum_\alpha \mu_{\alpha\alpha} + V(\vec{q}), \tag{2}$$

and our calculations are based on the symmetry-adapted canonical normal coordinates q_i. Surfaces such as the PES $V(\vec{q})$, the inverse moment of inertia tensor $\mu_{\alpha\beta}$, or the dipole moment surface (DMS) are represented by an n-mode expansion[26] and, thus, include terms of increasing dimensionality. Each of these terms can be considered a difference surface obtained from subtracting the lower-dimensional surfaces such that every term is a correction to the terms of preceding order. This means that the importance of a term decreases with the increase in dimensionality, usually ensuring fast convergence. Recent generalizations of our algorithms allow for an arbitrary order in the n-mode expansion within our programs although a truncation after the three-mode or four-mode coupling terms, usually, is fully sufficient for applications focusing on low-lying vibrational states,[31,32] as in the study here. Surfaces are calculated in a fully automated manner on a grid of *ab initio* points utilizing symmetry to reduce the number of single point calculations for arbitrary point groups.[33] As convergence of the n-mode expansion should be checked individually for the molecule of interest, this can easily be accomplished with our automated surface generator. For most applications, surfaces are subsequently fitted to obtain an analytical representation with different regression functions such as polynomials, B-splines, or distributed Gaussians. Due to the sum-of-products (SOP) representation of the PES and DMS, a very efficient Kronecker product fitting can be used.[34]

For the rovibrational calculations, purely vibrational wave functions are determined in a first step, which are subsequently used to obtain rovibrational energies and intensities. The vibrational wave functions are provided by VSCF and VCI calculations.[14,28] The former method uses a simple ansatz of a single Hartree product per state to describe the wave functions, $|\phi_v\rangle = |\prod_i \varphi(q_i)\rangle$. Different kinds of basis functions are implemented for the single-particle wave functions (modals) $\varphi(q_i)$, such as harmonic oscillator functions or distributed Gaussians. Each of these basis sets features a set of parameters that are varied iteratively, until self-consistency is achieved and quantities such as energies converge. The VCI method considers correlation by using an expansion including many configurations (Hartree products),

$$|\Phi_v\rangle = \sum_I c_I |\phi_v^I\rangle. \tag{3}$$

The vibrational wave functions $|\Phi_v\rangle$ are then used as basis functions for the rovibrational wave functions,

$$|\Psi_{\mathrm{rovib.}}\rangle = \sum_{rv} c_{rv}|\Phi_r\rangle|\Phi_v\rangle. \tag{4}$$

Rotational basis functions $|\Phi_r\rangle$ are chosen to be Wang combinations $|JK\tau\rangle$, with quantum numbers J, $0 \le K \le J$ and $\tau \in \{0, 1\}$, as introduced in Ref. 35,

$$|J0\tau\rangle = \begin{cases} |J0\rangle & \text{for } \tau = 0 \\ 0 & \text{for } \tau = 1, \end{cases} \tag{5a}$$

$$|JK\tau\rangle = \frac{i^\sigma}{\sqrt{2}}\left(|JK\rangle + (-1)^{J+K+\tau}|J-K\rangle\right) \text{ for } K \neq 0. \tag{5b}$$

Here, $|Jk\rangle$ are the common rigid rotor functions with $-J \le k \le J$, corresponding to the Wigner rotation matrix elements $D^J_{km}(\phi, \theta, \chi)$

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as functions of the Euler angles and $\sigma = 0$ for $\tau = 0$ and $\sigma = K \bmod 3$ for $\tau = 1$. The rotational quantum number m only plays a role in the presence of external fields and is neglected throughout this work. Hence, it is omitted, and energies are $(2J + 1)$-fold degenerate with respect to this quantum number.

To express the rotational part of the matrix elements more compactly, we introduce a further expansion, corresponding to the Wang combinations (5),

$$|JK\tau\rangle = \sum_{\tilde{\tau}=0}^{1} t_{\tilde{\tau}}^{JK\tau} \left|J(-1)^{\tilde{\tau}}K\right\rangle, \tag{6}$$

with

$$t_{\tilde{\tau}}^{JK\tau} = \frac{i^{\sigma}}{\sqrt{2}} \times \begin{cases} 1 & \text{for } \tilde{\tau} = 0 \\ (-1)^{J+K+\tau} & \text{for } \tilde{\tau} = 1. \end{cases} \tag{7}$$

Special care has to be taken in the case of $K = 0$, as in Eq. (5a).

With this, the rotational part of the matrix elements simply becomes

$$\left\langle J'K'\tau'\middle|H_{\text{rot}}\middle|J''K''\tau''\right\rangle = \sum_{\tilde{\tau}''\tilde{\tau}'} \left(t_{\tilde{\tau}}^{JK\tau}\right)'^{*} \left(t_{\tilde{\tau}}^{JK\tau}\right)'' \times \left\langle J'(-1)^{\tilde{\tau}'}K'\middle|H_{\text{rot}}\middle|J''(-1)^{\tilde{\tau}''}K''\right\rangle, \tag{8}$$

expressed again in the primitive rigid rotor states $|Jk\rangle$ for convenience. The matrix elements with respect to these basis functions have analytic results, and the only non-vanishing elements are[36]

$$\langle Jk|H_{\text{rot}}|Jk\rangle = \frac{1}{2}\left[(\mu_{xx} + \mu_{yy})\frac{1}{2}\left[J(J+1) - k^2\right] + \mu_{zz}k^2\right], \tag{9a}$$

$$\langle Jk \pm 1|H_{\text{rot}}|Jk\rangle = (\mu_{xz} \pm i\mu_{yz})\frac{1}{4}[k + k \pm 1]\sqrt{J(J+1) - k(k \pm 1)}, \tag{9b}$$

$$\langle Jk \pm 2|H_{\text{rot}}|Jk\rangle = -\frac{1}{8}(\mu_{xx} - \mu_{yy} \pm 2i\mu_{xy})\sqrt{J(J+1) - k(k \pm 1)} \times \sqrt{J(J+1) - (k \pm 1)(k \pm 2)}. \tag{9c}$$

Blocks for different values of J are independent such that the problem can be treated for one value at a time.

Note that we also consider a non-diagonal inverse moment of inertia tensor $\mu_{\alpha\beta}$, as well as the Coriolis coupling, dropping the widely used adiabatic rotation approximation (ARA).[37] This results in matrix elements on the off-diagonals $\Delta k = \pm 1$ from the angular momentum operators J_xJ_z, J_zJ_x, J_yJ_z, and J_zJ_y.

A connection can be established between the matrix eigenvalue problem in the basis of the primitive rigid rotor basis $\boldsymbol{H}^{\text{pr}}$ and the Wang functions $\boldsymbol{H}^{\text{Wang}}$. This provides the possibility of switching between the two representations, which can be advantageous as the Wang combinations feature a further blocking of the Hamiltonian matrix but lead to significantly more complex expressions for quantities such as transition moments. The Hamiltonian matrices in the two basis sets are connected by a similarity transformation,

$$\boldsymbol{H}_{\text{Wang}} = \boldsymbol{U}^{\dagger}\boldsymbol{H}_{\text{primitive}}\boldsymbol{U}, \tag{10}$$

with the transformation matrix U containing the Wang combination coefficients,

$$U_{0,0} = 1, \tag{11a}$$

$$U_{2K,\tau J+K} = i^{\sigma}/\sqrt{2}, \tag{11b}$$

$$U_{2K+1,\tau J+K} = (-1)^{J+K+\tau} i^{\sigma}/\sqrt{2}. \tag{11c}$$

As an example, for $J = 2$, one obtains

$$U = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{i}{\sqrt{2}} & 0 \\ 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad U^{\dagger} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}. \tag{12}$$

The rotational part of the Wang transformed Hamiltonian, expressed in matrix elements with respect to the rigid rotor basis (no superscript), is

$$\boldsymbol{H}^{\text{Wang}} = \begin{pmatrix} H_{0,0} & \frac{1}{\sqrt{2}}(H_{0,1} - H_{0,-1}) & \frac{1}{\sqrt{2}}(H_{0,2} + H_{0,-2}) & \frac{i}{\sqrt{2}}(H_{0,1} + H_{0,-1}) & \frac{-1}{\sqrt{2}}(H_{0,2} - H_{0,-2}) \\ \frac{1}{\sqrt{2}}(H_{1,0} - H_{-1,0}) & \frac{1}{2}(H_{1,1} + H_{-1,-1} - H_{1,1} - H_{1,-1}) & \frac{1}{2}(H_{1,2} - H_{-1,-2}) & 0 & 0 \\ \frac{1}{\sqrt{2}}(H_{2,0} + H_{-2,0}) & \frac{1}{2}(H_{2,1} - H_{-2,-1}) & \frac{1}{2}(H_{2,2} + H_{-2,-2}) & 0 & 0 \\ \frac{-i}{\sqrt{2}}(H_{1,0} + H_{-1,0}) & 0 & 0 & \frac{1}{2}(H_{1,1} + H_{-1,-1} + H_{-1,1} + H_{1,-1}) & \frac{i}{2}(H_{1,2} - H_{-1,-2}) \\ \frac{-i}{\sqrt{2}}(H_{2,0} - H_{-2,0}) & 0 & 0 & -\frac{i}{2}(H_{2,1} - H_{-2,-1}) & \frac{1}{2}(H_{2,2} + H_{-2,-2}) \end{pmatrix}. \tag{13}$$

Most notably, a block-structure emerges for $K' \neq 0$ and $K'' \neq 0$ as matrix elements between states of different quantum numbers τ vanish. A complete blocking is prevented by non-vanishing contributions in the first row and column, respectively, from terms of the form $J_\alpha \mu_{\alpha\beta} J_\beta$, $\alpha \neq \beta$ involving off-diagonal elements of the inverse moment of inertia tensor $\mu_{\alpha\beta}$. Within the adiabatic rotation approximation (ARA), these elements are neglected, which would lead to a matrix that is fully block-diagonal, and the diagonalization

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could be performed on the sub-blocks separately, reducing the computational effort. However, since, in our approach, the size of the vibrational basis is comparatively small, the size of the overall rovibrational matrix is moderate and diagonalization does not constitute a computational bottleneck.

The case of both $K' = 1$ and $K'' = 1$ is a special case as this is the only matrix element where the contributions $H_{-K',K''}$ and $H_{K',-K''}$ are not zero due to coupling by the angular momentum operators J_x^2, J_y^2, J_xJ_y and J_yJ_x; see Eq. (9c).

The vibrational part of the matrix elements cannot be expressed analytically and needs to be treated numerically. Employing the n-mode expansion of the inverse moment of inertia tensor $\mu_{\alpha\beta}$ and the potential, the integrals to be computed are quite numerous and expensive. Also, because of the truncated n-mode expansion, the inverse moment of inertia tensor $\mu_{\alpha\beta}$ no longer commutes with the vibrational angular momentum operators π_α, such that the simplification derived by Watson[27,38] cannot be used. The treatment of the vibrational part was discussed in a previous publication.[39]

For the incorporation of rotation, the Coriolis coupling terms $H_{\text{rovib.}} = -\frac{1}{2}\sum_{\alpha\beta}\left(J_\alpha\mu_{\alpha\beta}\pi_\beta + \pi_\alpha\mu_{\alpha\beta}J_\beta\right)$ had to be implemented, involving integrals of the following form:

$$\langle v'|\mu_{\alpha\beta}\pi_\beta|v''\rangle\langle JK'\tau'|J_\alpha|JK''\tau''\rangle. \tag{14}$$

Using the n-mode representation of the $\mu_{\alpha\beta}$ tensor and choosing functions to describe the surfaces and wave functions, one obtains integrals that are solvable analytically up to coefficients and parameters.

The Coriolis coupling matrix elements arising from the angular momentum operators J_x, J_y, and J_z with respect to the rigid rotor basis are given by[40]

$$\langle Jk|\hat{H}_{\text{cc}}|Jk\rangle = -\frac{1}{2}\sum_\alpha(\mu_{\alpha z}\pi_\alpha + \pi_\alpha\mu_{\alpha z})k, \tag{15a}$$

$$\langle Jk\pm 1|\hat{H}_{\text{cc}}|Jk\rangle = -\frac{1}{2}\left[\sum_\alpha(\mu_{\alpha x}\pi_\alpha + \pi_\alpha\mu_{\alpha x}) \mp i\sum_\alpha(\mu_{\alpha y}\pi_\alpha + \pi_\alpha\mu_{\alpha y})\right] \times \frac{1}{2}\sqrt{J(J+1) - k(k\pm 1)}. \tag{15b}$$

By including these matrix elements, the Hamiltonian matrix for the rovibrational problem involves couplings between different vibrational states corresponding to off-diagonal sub-matrices. The total matrix for a given value of J reads

$$\boldsymbol{H} = \begin{pmatrix} \begin{pmatrix} H_{\text{vib}}^{00} + H_{\text{rot}}^{00} & H_{\text{rot}}^{01} & \cdots \\ H_{\text{rot}}^{10} & H_{\text{vib}}^{00} + H_{\text{rot}}^{11} & \\ \vdots & & \ddots \end{pmatrix} & \begin{pmatrix} H_{\text{cc}}^{00,10} + H_{\text{rot}}^{00} & H_{\text{cc}}^{00,11} + H_{\text{rot}}^{01} & \cdots \\ H_{\text{cc}}^{01,10} + H_{\text{rot}}^{10} & H_{\text{cc}}^{01,11} + H_{\text{rot}}^{11} & \\ \vdots & & \ddots \end{pmatrix} & \cdots \\ \begin{pmatrix} H_{\text{cc}}^{10,00} + H_{\text{rot}}^{00} & H_{\text{cc}}^{10,01} + H_{\text{rot}}^{01} & \cdots \\ H_{\text{cc}}^{11,00} + H_{\text{rot}}^{10} & H_{\text{cc}}^{11,01} + H_{\text{rot}}^{11} & \\ \vdots & & \ddots \end{pmatrix} & \begin{pmatrix} H_{\text{vib}}^{11} + H_{\text{rot}}^{00} & H_{\text{rot}}^{01} & \cdots \\ H_{\text{rot}}^{10} & H_{\text{vib}}^{11} + H_{\text{rot}}^{11} & \\ \vdots & & \ddots \end{pmatrix} & \\ \vdots & & \ddots \end{pmatrix}. \tag{16}$$

Purely vibrational contributions $H_{\text{vib}}^{v'v''}$ occur only on the main diagonal of the overall matrix as they vanish for different vibrational states, $v' \neq v''$, and are naturally diagonal with respect to the vibrational wave functions. Rotational contributions $H_{\text{rot}}^{i_r'i_r''}$ [where i_r abbreviates (K, τ)] occur, as given in Eq. (13), with sparsity, due to the restriction $|K' - K''| \leq 2$ and a general block-structure for $\tau' \neq \tau''$, except for the first column and row. Coriolis coupling terms $H_{\text{cc}}^{i_v'i_r',i_v''i_r''}$ are strictly off-diagonal with respect to vibrational wave functions and give contributions to the main diagonal and two off-diagonals above and below with respect to rotational wave functions. A more complete graphical representation is given in Fig. 1.

The size of the matrix is $((2J+1)N_{\text{vib}})^2$, where N_{vib} denotes the number of vibrational states considered. Several properties could be used to reduce the computational effort in the diagonalization. The limitation in the rotational quantum numbers to $|\Delta k| \leq 2$ results in a significant sparsity. In addition, using the symmetrized basis functions of Wang combinations leads to a further (mostly) block-diagonal structure where matrix elements for different quantum numbers τ vanish. The purely vibrational contributions to the matrix elements are diagonal with respect to vibrational states. The Coriolis coupling terms, on the other hand, vanish in this case and only contribute to off-diagonals with respect to the vibrational basis; see Eq. (15b).

For the energy and temperature regime in our usual applications, rotational excitations up to $J_{\max} = 40$–50 have to be considered, and the vibrational ground state, fundamentals, and first excited vibrational states are necessary to have a sufficiently large vibrational basis set. As the VCI solutions are chosen as basis functions for the rovibrational problem, the size of the rovibrational matrix is usually not problematic as most of the computational cost is shifted to the calculation of the matrix elements. For these, even using our configuration selection schemes, a large number of configurations may have to be taken into account. The diagonalization of the resulting rovibrational Hamiltonian matrix is feasible with standard LAPACK libraries. Results are discussed later on in the context of intensities and spectra.

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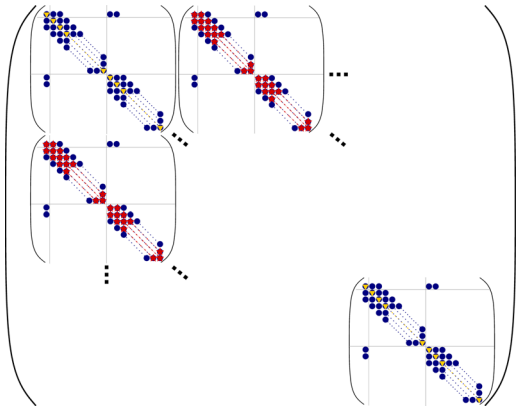

FIG. 1. Graphical representation of the rovibrational configuration interaction (RVCI) matrix. Blue dots, yellow triangles, and red stars represent contributions from the rotational, vibrational, and the Coriolis coupling term of the Watson operator, respectively. Gray lines divide the first column and row and the sub-blocks of different rotational quantum numbers τ that vanish if both $K_a'' \neq 0$ and $K_a' \neq 0$.

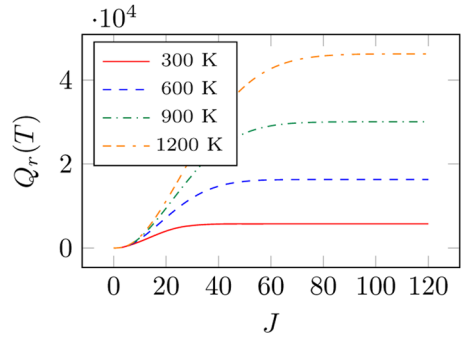

FIG. 2. Convergence of the rotational partition function with an increasing rotational quantum number J at different temperatures for thioformaldehyde. The maximum value to converge the partition function for a temperature of 1200 K is in this example, J_{max} = 120.

B. Partition functions

To avoid full energy calculations for highly excited states, necessary for converged partition functions, the approximation of separating vibrational and rotational contributions is used. This is well established[41] and results in errors usually fairly below those of other quantities contributing to the intensities. Consequently, the partition function can be approximated as follows:

$$Q = Q_{\text{vib}} Q_{\text{rot}}, \tag{17a}$$

with

$$Q_{\text{vib}} = \sum_v e^{-E_v/k_B T}, \quad Q_{\text{rot}}(T) = \sum_r g_r e^{-E_r/k_B T}, \tag{17b}$$

where $g_r = g_{\text{ns}}(2J + 1)$ are the degeneracies for a given rotational state r with respect to nuclear spin states and the rotational quantum number m. For the rotational part, the Hamiltonian matrix described in Sec. II A is built for increasing values of J assuming only the equilibrium structure rotational constants, corresponding to the 0D integrals of the inverse moment of inertia tensor $\mu_{\alpha\beta}$. Investigations have shown that higher order contributions have a negligible impact for thioformaldehyde, with less than 1% almost exclusively arising from 1D integrals, independent of the temperature. Hence, these terms have been neglected in the following. The resulting matrix is diagonalized to obtain the rotational energies, and Q_{rot} is updated until deviations are below a convergence threshold. In addition, nuclear spin statistical weights, as described in Sec. II C, have to be taken into account. Figure 2 shows the convergence for different temperatures where excitations of up to $J_{\text{max}} \approx 120$ were considered. The construction and diagonalization of the corresponding rotational Hamiltonian matrix are not time-critical.

To avoid the costly computation of highly excited vibrational states, those that are not already included in the current calculation are approximated using VSCF modal energies, i.e., the state energy is reduced to a sum of modal energies,

$$H_{\text{vib}}|\Phi_{\text{vib}}\rangle \approx H_{\text{vib}}\left|\prod_i \phi_i^{n_i}\right\rangle, \tag{18a}$$

$$E_v \approx \sum_i E_i^{n_i}. \tag{18b}$$

With this, a sufficiently good guess for very high overtones and combination bands is possible, which is again performed until convergence is achieved. As vibrational energies generally increase quite quickly with increasing excitation, only a relatively small number of states has to be taken into account. Fundamentals will make up for most of the contributions such that the approximation for higher excited states has a small impact. To increase accuracy, nonetheless, more vibrational states can be properly computed.

The error introduced by using the approximation of separating rotational and vibrational partition functions and arising from other approximations is within margins of less than one percent and is, thus, well within our desired accuracy, exceeded by other uncertainties. Table I shows values of the partition function for H_2CO and H_2CS at different temperatures compared with the experimental values of the JPL database[42] and of Maeda *et al.*[43] Note that the slightly newer experimental values of Maeda *et al.*[43] match our computed values better than those of the JPL database. The comparison of the two experimental datasets reveals the uncertainty within the experimental data, which is in the same range as the uncertainty of our computed values.

The direct product approximation can be expected to perform worse for increasing temperatures where more energy levels significantly contribute to the partition function as arising in astrochemical applications. Low-frequency vibrational motions of heavy molecules and molecular clusters will pose a problem as well, as overtones and combination bands will have to be calculated at high accuracy to obtain meaningful values of the vibrational partition function. In addition, larger and heavier molecules will show much slower convergence of the rotational partition function with increasing J, and in the high-J regime, the rigid rotor approximation might not perform well, just as for less rigid molecules.

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TABLE I. Partition function for H_2CO and H_2CS at different temperatures, computed and experimental values.

	H_2CO			H_2CS				
T/K	Calc.	Expt.[a]	Δ (%)	Calc.	Expt.[b]	Δ%	Expt.[c]	Δ (%)
75	362.1	362.1	0.0	743.2	750.1	−0.9	750.1	−0.9
150	1026.0	1020.5	0.5	2104.2	2088.8	0.7	2119.2	0.7
225	1887.0	1875.7	0.6	3883.6	3761.0	3.3	3893.3	0.2
300	2923.4	2906.3	0.6	6096.4	5984.1	1.9	5996.0	1.7
500	6802.3	6761.0	0.6	15 088.1	...	...		

[a]Taken from Ref. 44.
[b]Taken from Ref. 42.
[c]Taken from Ref. 43.

C. Nuclear spin statistical weights

The nuclear spin statistical weights g_{ns} are necessary to account for the degeneracy of energies with respect to different nuclear spin states. In this work, we employ the Landau–Lifshitz-formula,[45,46]

$$\chi_{rv}^{sw}[R] = 2\sum_a (2I_a+1)(-1)^{2I_a(n_a-1)}, \tag{19}$$

for the characters $\chi_{rv}^{sw}[R]$ of the nuclear spin *representation* Γ_{rv}^{sw} which is, in general, reducible. The index a denotes atom sets that contain symmetry-equivalent atoms for each symmetry operation R, I_a is the corresponding nuclear spin, and n_a the number of nuclei in a set. For the identity operation E, leaving all atoms unchanged, each nucleus is in its own set. The character for permutation-inversions is 0, as these do not affect the total spin wave function.[45,47] To obtain the actual statistical weights, this *representation* is reduced by standard means of group theory. The multiplicity of each irreducible representation is given by

$$a_{\Gamma_i} = \frac{1}{g}\sum_R \chi_{rv}^{sw}[R]\chi^i[R]^*, \tag{20}$$

with $\chi^i[R]$ denoting the character of symmetry operation R from the character table for a given irreducible representation i of the molecular symmetry group.

Consequently, nuclear spins and information about the atom sets or symmetry operations are needed. The former can be calculated in a few special cases of nucleus compositions, but generally, have to be entered via tables or user input. The latter could be realized in the form of tables as well, but are not general to one point group and differ for molecules with varying numbers of atoms. They are, for example, not the same for H_2O and H_2CO. As a consequence, it is desirable to have a general algorithm for identifying the atom sets a and the number of atoms per set n_a. As a similar analysis is performed in our surface generation program to determine the point group of the molecule, information about atom sets can be obtained by slight extensions of the existing code. Determination of the nuclear spin statistical weights in our program is currently limited to selected Abelian point groups, and a generalization to arbitrary symmetries is an ongoing work.

Table II illustrates the process of determining the characters and the irreducible representation with multiplicities corresponding to the nuclear spin statistical weights for H_2CO, H_2CS, and molecules of the same point group and nuclear spins. As a result, the nuclear spin statistical weights for transitions between $A_1 \leftrightarrow A_2$ states are three times as intense as those between $B_1 \leftrightarrow B_2$.

To deduce the nuclear spin wave function symmetry from the symmetry of the rovibrational wave function finally, the total internal wave function symmetry is required. In the case of Abelian groups, it can readily be obtained from the direct product,

$$\Gamma_{tot} = \Gamma_{ns} \otimes \Gamma_{rv}. \tag{21}$$

Here, Γ_{tot}, Γ_{ns}, and Γ_{rv} denote the total internal, the nuclear spin, and the rovibrational wave function symmetry, respectively. In this example, Γ_{tot} can be obtained from a simple analysis of even or odd permutations of fermions to be of the B type. The symmetry of the rovibrational wave function results from the combined symmetries of rotational and vibrational parts and in the case of an Abelian group can again be obtained from a simple direct product,

$$\Gamma_{rv} = \Gamma_{vib} \otimes \Gamma_{rot}. \tag{22}$$

The vibrational irrep Γ_{vib} is central information in the VSCF and VCI calculations and is known. For the rotational symmetry, the so-called equivalent rotations have to be determined, which correspond to the symmetry operations.[40] Applying these onto the rotational wave function reveals the transformation properties and such characters and allows for the assignment of the irrep, depending on the rotational quantum numbers. For thioformaldehyde, the correspondence is given in Table III.[47]

TABLE II. Character table for the C_{2v} point group and characters of the nuclear spin representation for formaldehyde-like molecules. The multiplicities of the associated irreducible representation correspond to the nuclear spin statistical weights. $\Gamma_{rve}^{sw} \overset{(20)}{\Rightarrow} \tilde{\Gamma}_{rve}^{sw} = A_1 \oplus A_2 \oplus 3B_1 \oplus 3B_2$

C_{2v}	E	(12)	E*	(12)*	g_{ns}
A_1	1	1	1	1	1
A_2	1	1	−1	−1	1
B_1	1	−1	−1	1	3
B_2	1	−1	1	−1	3
$\tilde{\Gamma}_{rve}^{sw}$	8	−4	0	0	

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TABLE III. Correspondence of rotational quantum numbers K_a, K_c to the symmetry of the rotational wave function in dependence of the rotational quantum numbers for molecules such as H_2CS and H_2CO. Here, the information whether the quantum numbers are even (e) or odd (o) is sufficient.

K_aK_c	ee	eo	oe	oo
Γ_{rot}	A_1	A_2	B_2	B_1

Knowledge about the rovibrational symmetry and the symmetry of the total internal wave function finally enables the assignment of the nuclear spin wave function symmetry and, hence, the correct nuclear spin statistical weight.

D. Transition moments

Finally, transition moments or line strengths are needed to obtain intensities. To this end, integrals of the general form $\langle \Psi' | d | \Psi'' \rangle$ have to be determined, with the wave functions Ψ', Ψ'' of the initial and final states, respectively, and the dipole operator d. We restrict transitions to rovibrational states and do not consider, for example, electronic excitations. The derivation of rovibrational transition moments is commonly known and can be found, for example, in the textbook of Bunker and Jensen.[47] With the modification of using the Wang combinations (IIA) and expressing them again in the primitive basis of the rigid rotor states (7), the line strengths become

$$R = g_{\mathrm{ns}}(2J''+1)(2J'+1)\left|\sum_{\sigma=-1}^{1}(-1)^{\sigma}\sum_{v''v'}\langle v'|d_{\mathrm{m}}^{(1,\sigma)}|v''\rangle \times \sum_{\substack{K''K'\\ \tau''\tau'}}(-1)^{K''}(c_{vK\tau})'^{*}(c_{vK\tau})'' \times \sum_{\tilde{\tau}''\tilde{\tau}'}\left(t_{\tilde{\tau}}^{JK\tau}\right)'^{*}\left(t_{\tilde{\tau}}^{JK\tau}\right)''\begin{pmatrix} J'' & 1 & J' \\ (-1)^{\tilde{\tau}''}K'' & \sigma & (-1)^{\tilde{\tau}'+1}K' \end{pmatrix}\right|^{2}. \tag{23}$$

The spherical components of the molecule-fixed dipole tensor $d_{\mathrm{m}}^{(1,\sigma)}$ are connected to the Cartesian components as $d_{\mathrm{m}}^{(1,\pm1)} = \mp d_x + i d_y$, $d_{\mathrm{m}}^{(1,0)} = d_z$, and the $3J$-symbol is zero unless $(-1)^{\tilde{\tau}''}K'' - (-1)^{\tilde{\tau}'}K' = \sigma$ and $|J''-J'| \leq 1$.[40] Degeneracies with respect to the rotational quantum number m and nuclear spin states have to be taken into account, leading to prefactors $g_{\mathrm{ns}}(2J''+1)\,(2J'+1)$.

A major challenge in generating line lists is the correct assignment of quantum numbers to the states involved in a transition. In principle, the assignment can be performed by considering the position of the leading coefficient in the eigenvector of the Hamiltonian matrix. Since the order of the basis vectors is known, every position in the eigenvector corresponds to one basis function. The assignment gets challenging when the leading coefficients are ambiguous. To overcome this problem, Wang functions are used. This leads to a significant reduction in ambiguous cases from about 30% with trivial rotation basis to about 3% using Wang combinations. However, there are still cases where the quantum numbers cannot be assigned correctly. For this reason, the basis functions of the final rovibrational matrix are further refined. This is performed by first solving the pure rotational problem for the vibrational ground state in the Wang basis. The resulting eigenvectors are used as a new molecule specific rotational basis (MSRB). We introduced this extra step within the algorithm because it takes hardly any time, but stabilizes the calculation significantly. With this basis, the state identity is unambiguous up to about $J_{\max} = 40$ for fundamental transitions. For vibrational overtones and combination bands, the situation changes and the ambiguity increases, but, of course, much less than without the MSRB. With this, the transition moments are given as follows:

$$R = g_{\mathrm{ns}}(2J''+1)(2J'+1)\left|\sum_{\sigma=-1}^{1}(-1)^{\sigma}\sum_{v''v'}\langle v'|d_{\mathrm{m}}^{(1,\sigma)}|v''\rangle \times \sum_{n''n'}(c_{vn})'^{*}(c_{vn})'' \sum_{\substack{K''K'\\ \tau''\tau'}}(-1)^{K''}\left(c_{K\tau}^{n}\right)'^{*}\left(c_{vK\tau}^{n}\right)'' \times \sum_{\tilde{\tau}''\tilde{\tau}'}\left(t_{\tilde{\tau}}^{JK\tau}\right)'^{*}\left(t_{\tilde{\tau}}^{JK\tau}\right)''\begin{pmatrix} J'' & 1 & J' \\ (-1)^{\tilde{\tau}''}K'' & \sigma & (-1)^{\tilde{\tau}'+1}K' \end{pmatrix}\right|^{2}. \tag{24}$$

Here, c_{vn} corresponds to the MSRB coefficients, $c_{K\tau}^{n}$ describes the rovibrational coefficient, and $t_{\tilde{\tau}}^{JK\tau}$ is assigned to the Wang coefficients.

Transitions are limited by the common selection rule $|J''-J'| \leq 1$ and to $A_1 \leftrightarrow A_2$ or $B_1 \leftrightarrow B_2$ by symmetry considerations. While this reduces the number of significant transitions in the spectrum tremendously, there can still be an enormous number of lines that have to be taken into account. This is especially true for heavier and larger molecules, where energies increase more slowly with excitation than for lighter molecules, as discussed above, in the context of partition functions. Also, for high-temperature applications, the number of states that have to be taken into account may increase drastically, easily reaching millions to billions, up to a quasi-continuum. To address this, special basis sets and reduction to symmetry sub-spaces may be used to reduce the effort for determining line intensities. Cutoffs and data compression have to be introduced, while maintaining all relevant features and the total integrated intensity. Concepts such as combining multiple features into super-lines help handling the large amount of data.[48–52]

For the energy and temperature regimes investigated within this work, the number of transitions is moderate though, such that introducing an intensity threshold is sufficient to handle the amount of data. Also, no special care is taken regarding the vibrational and rotational bases beside a limitation of the latter to fundamental modes and first overtones and combination bands.

E. Voigt profile

For a physical description of the line profiles, Doppler and pressure broadening are taken into account. These result in a combined Gaussian and a Lorentzian line shape in the form of a Voigt profile.[53,54] This is achieved by the convolution of a Gauss function with a Lorentz function,

$$V(\nu;\sigma,\gamma) = \int G(\tilde{\nu};\sigma)L(\nu-\tilde{\nu};\gamma)\,\mathrm{d}\tilde{\nu}, \tag{25}$$

with the temperature (T) and pressure (P) dependent parameters σ and γ connected to the half widths of the Gauss and Lorentz functions, respectively,

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$$\sigma = \frac{2}{c}\sqrt{\frac{2\ln 2 k_B T}{m}}, \quad \gamma = \frac{Pr_{\text{char}}^2}{c}\sqrt{\frac{2}{m k_B T}}, \quad a = \gamma/\sqrt{2}\sigma. \tag{26}$$

Using a allows for a representation with only one parameter. In the limit $a \gg 1$, the Voigt profile is dominantly Gauss shaped, and for $a \ll 1$, the Lorentz shape is more prominent. The variable c denotes the velocity of light, k_B is the Boltzmann constant, m is the molecule mass, and r a characteristic radius of the molecule.

As there is no analytical solution to the convolution needed for the Voigt profile, the integral is evaluated numerically. With an increasing number of features in the spectrum and a high sample rate, the computation of the overall spectrum becomes computationally expensive. Various technical refinements are employed to increase the efficiency. For a given reference frequency, corresponding to a single line in the spectrum, the Voigt profile is computed only within a window up to a point where the function is sufficiently small. Symmetry of the Voigt profile is used such that only one half is calculated and then mirrored. Approximate methods exist to estimate the Voigt profile using, for example, the pseudo-Voigt function. Within this work, the *exact* method of calculating the convolution was used though.

III. BENCHMARK CALCULATIONS

As thioformaldehyde is sufficiently small, shows strongly pronounced Coriolis coupling effects, and has extensively been studied by several authors, it serves as a benchmark system for rovibrational calculations.[20,30,43,55–60] Due to these couplings and the occurrence of two quasi-degenerate vibrational frequencies, pronounced intensity borrowing can be observed, rendering this spectrum fairly sensitive with respect to any underlying approximations. The fundamental vibrational frequencies are shown in Table IV. The overall agreement of our values with those obtained from multi-configuration time-dependent Hartree (MCTDH) calculations,[55] those of Yachmenev *et al.*,[30] and the experimental gas phase data, as taken from the NIST database,[61] is excellent. With respect to the vibrational overtones, the MCTDH calculations fail in a proper description of the strongly coupling states $2\nu_4$ and $2\nu_6$. We will focus, in particular, on modes ν_4 and ν_6, which, due to their quasi-degeneracy, give rise to the strong Coriolis coupling effects in the rovibrational calculations as mentioned above.

TABLE IV. Fundamental transitions and first vibrational overtones of thioformaldehyde.

		Energy (cm^{-1})			
State	Sym.	This work	TROVE[a]	MCTDH[b]	Expt.[c]
ν_4	B_1	989.4	990.6	990.5	990.2
ν_6	B_2	989.5	991.0	991.0	991.0
ν_3	A_1	1060.2	1060.6	1058.9	1059.2
ν_2	A_1	1454.8	1455.7	1457.4	1455.5
ν_1	A_1	2969.1	2972.4	2970.0	2971.0
ν_5	B_2	3024.7	3027.6	3024.5	3024.6
$2\nu_6$	A_1	1964.9	1967.1	1979.1	
$2\nu_4$	A_1	1990.8	1992.9	1939.5	
$2\nu_3$	A_1	2110.7	2111.5	2101.8	
$2\nu_2$	A_1	2875.4	2877.8	2876.0	2877.1
$2\nu_1$	A_1	5796.4		5789.0	
$2\nu_5$	A_1	6009.3		6013.8	

[a]Taken from Ref. 30.
[b]Taken from Ref. 55.
[c]Taken from Ref. 61.

A. Computational details

The geometry and harmonic frequencies of H_2CS have been determined at the CCSD(T)-F12a/cc-pVTZ-F12 level.[62] The frozen core approximation has been used in all calculations. The equilibrium geometry of thioformaldehyde has been taken as the expansion point of an n-mode expansion by means of normal coordinates up to the fourth order.[26,28] A multi-level scheme[63] has been employed, in which a cc-pVTZ-F12 basis set has been used for the 1D and 2D surfaces and a cc-pVDZ-F12 basis for 3D and 4D terms. Dipole moment surfaces (DMSs) were obtained from conventional coupled-cluster [CCSD(T)] single-point calculations including the coupled-perturbed Hartree–Fock equations. Both, the PES and DMS, which can be downloaded from our database of potential energy surfaces,[64] were determined in a fully automated fashion exploiting symmetry within the electronic structure calculations and the individual subsurfaces of the n-mode expansion.[33] Due to the iterative build-up of the surfaces, convergence in all regions of the PES and DMS is guaranteed. In a subsequent step efficient Kronecker product fitting[34] has been used to transform the grid representation into an analytical sum-of-products representation.

Vibrational self-consistent field theory (VSCF) based on the Watson Hamiltonian[27] has been used to determine variationally optimized one-mode wave functions (modals). A basis of 20 distributed Gaussians has been used within these initial calculations. Ground-state based modals have been employed to generate a basis of Hartree products (configurations) for the subsequent configuration selective vibrational configuration interaction (VCI) calculations. Six-tuple excitations, a sum of quantum numbers of 15, and a maximum excitation up to the seventh root for one mode were used to restrict the configuration space. Wang combinations as described above [cf. Eq. (5)] were then used as rotational basis functions to solve the purely rotational problem, resulting in the molecule specific rotational basis (MSRB, see Sec. II D).

Rovibrational calculations were then performed using the VCI wave functions as vibrational basis and the MSRB wave functions as rotational basis, resulting in the rovibrational configuration interaction (RVCI) matrix as given in Eq. (16). The correlation space included vibrational fundamentals and all rotational states for a given value of J. Integrals of the inverse moment of inertia tensor $\mu_{\alpha\beta}$ were evaluated up to the 2D terms in the n-mode expansion. This concerns the off-diagonal rotational terms,

$$\langle \nu' | \mu_{\alpha\beta} | \nu'' \rangle \langle JK'\tau' | J_\alpha J_\beta | JK''\tau'' \rangle, \tag{27}$$

and the Coriolis contributions,

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$$\langle v'|\mu_{\alpha\beta}\pi_\beta|v''\rangle\langle JK'\tau'|J_\alpha|JK''\tau''\rangle. \tag{28}$$

Careful assignment of the vibrational and rotational quantum numbers allowed for the application of the corresponding nuclear spin statistical weights. In order to speed up the calculation, transition moments were evaluated, skipping terms within the sums, where one of the wave function coefficients c_{vn} in Eq. (24) was below a certain threshold. Temperature effects were considered by applying the Boltzmann factors and partition function as shown in Eq. (1). For a temperature of T = 300 K, the vibrational partition function converged to less than 10^{-4} using the computed VCI vibrational energies and rotational excitations up to J = 54. All calculations were performed within the MOLPRO program package.[29]

B. Line list for thioformaldehyde

For thioformaldehyde (H_2CS), the nuclear spin statistical weights are 1 for transitions between rovibrational states with symmetries $A_1 \leftrightarrow A_2$ and 3 for transitions $B_1 \leftrightarrow B_2$ (see Table II). The partition function for a temperature of 300 K is determined to be 6096.4 in comparison to an experimental reference value of 5996.0.[43] In Tables VI and VII, we have listed a selection of rovibrational transitions for H_2CS and compare them with the experimental data and those computed by Yachmenev *et al.*[30] As can be seen, the transition energies are in excellent agreement with both the experimental and computational reference data. Deviations were found to be less than 2 cm^{-1}. This remaining error arises mainly from core correlation effects, high-order coupled-cluster terms, relativistic contributions, basis set deficiencies, and the neglect of diagonal Born–Oppenheimer corrections within the electronic structure calculations. Note that the potential energy surface of Yachmenev *et al.* accounts for these effects and has been further corrected to match the experimental results best, while our PES is just a simple uncorrected coupled-cluster surface. The reason for using a conventional *ab initio* surface is to show the accuracy obtainable from standard procedures without further refinements and applicability to larger systems. In principle, every calculation in this paper should be scalable to larger systems with feasible computational effort. Consequently, differences have to be anticipated, but were found to be fairly small. It can be seen from Tables V–VII that the agreement of the rovibrational intensities computed by our approach is excellent for many transitions (cf. Table V), while for others (cf. Table VI), it

TABLE V. Line list for the molecule thioformaldehyde for the ν_3 transitions at 1060.21 cm^{-1} at a temperature of 300 K, resulting in a partition function of Q(300 K) = 6096.4. The point group is C_{2v}, and the nuclear spin statistical weights are 1 for transitions $A_1 \leftrightarrow A_2$ and 3 for transitions $B_1 \leftrightarrow B_2$. Intensities are given in cm^{-1}/(molecule cm^{-2}).

J''	K_a''	K_c''	E''	$E''_{Calc.}$ [a]	$E''_{Expt.}$ [b]	J'	K_a'	K_c'	f	$f_{Calc.}$ [a]	$f_{Expt.}$ [b]	I	$I_{Calc.}$ [a]	$I_{Expt.}$ [b]
0	0	0	0.00	0.0000	0.0000	1	0	1	1061.31	1061.7210	1060.3450	2.253×10^{-22}	2.360×10^{-22}	2.293×10^{-22}
1	0	1	1.15	1.1458	1.1458	0	0	0	1059.02	1059.4349	1058.0589	2.226×10^{-22}	2.340×10^{-22}	2.296×10^{-22}
1	0	1	1.15	1.1458	1.1458	2	0	2	1062.46	1062.8556	1061.4796	4.497×10^{-22}	4.701×10^{-22}	4.558×10^{-22}
1	1	1	10.35	10.2819	10.2818	1	1	0	1060.20	1060.6016	1059.2251	9.634×10^{-22}	1.012×10^{-21}	9.811×10^{-22}
1	1	1	10.35	10.2819	10.2818	2	1	2	1062.42	1062.8151	1061.4390	9.763×10^{-22}	1.001×10^{-21}	9.742×10^{-22}
1	1	0	10.39	10.3170	10.3167	1	1	1	1060.13	1060.5328	1059.1566	9.562×10^{-22}	1.010×10^{-21}	9.916×10^{-22}
1	1	0	10.39	10.3170	10.3167	2	1	1	1062.49	1062.8813	1061.5049	9.833×10^{-22}	1.003×10^{-21}	9.637×10^{-22}
2	0	2	3.44	3.4373	3.4374	1	0	1	1057.87	1058.2837	1056.9076	4.388×10^{-22}	4.622×10^{-22}	4.558×10^{-22}
2	0	2	3.44	3.4373	3.4374	3	0	3	1063.60	1063.9844	1062.6084	6.694×10^{-22}	6.985×10^{-22}	6.715×10^{-22}
2	1	2	12.61	12.5384	12.5385	1	1	1	1057.90	1058.3113	1056.9348	9.500×10^{-22}	9.957×10^{-22}	9.776×10^{-22}
2	1	2	12.61	12.5384	12.5385	2	1	1	1060.26	1060.6598	1059.2831	5.240×10^{-22}	5.650×10^{-22}	5.427×10^{-22}
2	1	2	12.61	12.5384	12.5385	3	1	3	1063.55	1063.9293	1062.5534	1.724×10^{-21}	1.757×10^{-21}	1.705×10^{-21}
2	1	1	12.71	12.6436	12.6432	1	1	0	1057.84	1058.2400	1056.8637	9.426×10^{-22}	9.931×10^{-22}	9.846×10^{-22}
2	1	1	12.71	12.6436	12.6432	2	1	2	1060.06	1060.4534	1059.0776	5.121×10^{-22}	5.612×10^{-22}	5.567×10^{-22}
2	1	1	12.71	12.6436	12.6432	3	1	2	1063.65	1064.0267	1062.6502	1.742×10^{-22}	1.762×10^{-21}	1.684×10^{-21}
2	2	1	40.33	40.0423	40.0417	2	2	0	1060.16	1060.5323	1059.1549	6.153×10^{-22}	6.496×10^{-22}	6.332×10^{-22}
2	2	1	40.33	40.0423	40.0417	3	2	2	1063.60	1063.9441	1062.5681	2.392×10^{-22}	3.830×10^{-22}	3.385×10^{-22}
2	2	0	40.33	40.0424	40.0418	2	2	1	1060.16	1060.5321	1059.1547	6.153×10^{-22}	6.496×10^{-22}	6.332×10^{-22}
2	2	0	40.33	40.0424	40.0418	3	2	1	1063.60	1063.9445	1062.5684	2.392×10^{-22}	3.830×10^{-22}	3.385×10^{-22}
3	0	3	6.88	6.8743	6.8745	2	0	2	1056.72	1057.1271	1055.7510	6.452×10^{-22}	6.808×10^{-22}	6.750×10^{-22}
3	1	3	16.01	15.9231	15.9234	2	1	2	1056.77	1057.1739	1055.7974	1.654×10^{-22}	1.744×10^{-21}	1.712×10^{-21}
3	1	3	16.01	15.9231	15.9234	3	1	2	1060.36	1060.7471	1059.3700	3.553×10^{-22}	3.984×10^{-22}	3.757×10^{-22}
3	1	2	16.21	16.1333	16.1329	2	1	1	1056.67	1057.0649	1055.6887	1.635×10^{-22}	1.737×10^{-21}	1.733×10^{-21}
3	1	2	16.21	16.1333	16.1329	3	1	3	1059.96	1060.3344	1058.9590	3.392×10^{-22}	3.933×10^{-22}	3.931×10^{-22}
3	2	2	43.78	43.4793	43.4787	2	2	1	1056.72	1057.0952	1055.7178	3.019×10^{-22}	3.168×10^{-22}	3.128×10^{-22}
3	2	2	43.78	43.4793	43.4787	3	2	1	1060.16	1060.5076	1059.1315	4.722×10^{-22}	4.107×10^{-22}	4.175×10^{-22}
3	2	1	43.78	43.4798	43.4792	2	2	0	1056.72	1057.0948	1055.7174	3.018×10^{-22}	3.168×10^{-22}	3.128×10^{-22}
3	2	1	43.78	43.4798	43.4792	3	2	2	1060.16	1060.5066	1059.1305	4.721×10^{-22}	4.107×10^{-22}	4.175×10^{-22}
3	3	1	89.89	89.2000	89.1985	3	3	0	1060.16	1060.4758	1059.0967	2.291×10^{-22}	2.427×10^{-21}	2.355×10^{-21}
3	3	0	89.89	89.2000	89.1985	3	3	1	1060.16	1060.4758	1059.0967	2.291×10^{-22}	2.427×10^{-21}	2.355×10^{-21}
rms[c]			0.24	0.0005					1.00	1.377		7.351×10^{-22}	3.208×10^{-23}	

[a]Taken from Ref. 30.
[b]Taken from Ref. 65.
[c]Root mean square deviation (calc-obs).

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TABLE VI. Line list for the molecule thioformaldehyde for the ν_4 and ν_6 transitions at 989.36 cm^{-1} and 989.47 cm^{-1} at a temperature of 300 K, resulting in a partition function of Q(300 K) = 6096.4. The point group is C_{2v}, and the nuclear spin statistical weights are 1 for transitions $A_1 \leftrightarrow A_2$ and 3 for transitions $B_1 \leftrightarrow B_2$. Intensities are given in cm^{-1}/(molecule cm^{-2}).

J''	K''_a	K''_c	E''	$E''_{\text{Calc.}}$[a]	$E''_{\text{Expt.}}$[b]	ν'	J'	K'_a	K'_c	f	$f_{\text{Calc.}}$[a]	$f_{\text{Expt.}}$[b]	I	$I_{\text{Calc.}}$[a]	$I_{\text{Expt.}}$[b]
0	0	0	0.00	0.0000	0.0000	4/6	1	1	0	989.95	991.09	990.89	7.538×10^{-22}	1.249×10^{-21}	1.483×10^{-21}
0	0	0	0.00	0.0000	0.0000	4/6	1	1	1	1009.71	1011.08	1010.93	3.324×10^{-22}	6.619×10^{-22}	5.095×10^{-22}
1	0	1	1.15	1.1458	1.1458	4/6	2	1	1	991.09	992.22	992.02	1.125×10^{-21}	1.853×10^{-21}	2.185×10^{-21}
1	0	1	1.15	1.1458	1.1458	4/6	2	1	2	1010.85	1012.22	1012.06	4.940×10^{-22}	9.964×10^{-22}	7.678×10^{-22}
1	1	1	10.36	10.2819	10.2818	4	2	0	2	982.70	984.12	984.17	6.188×10^{-23}	5.493×10^{-23}	1.412×10^{-22}
1	1	1	10.36	10.2819	10.2818	4/6	2	2	0	1038.95	1040.56	1040.41	1.431×10^{-21}	2.886×10^{-21}	2.325×10^{-21}
1	1	1	10.36	10.2819	10.2818	4/6	2	2	1	999.44	1000.59	1000.37	3.260×10^{-21}	5.432×10^{-21}	6.508×10^{-21}
1	1	0	10.39	10.3170	10.3167	4	0	0	0	979.07	980.27	979.87	2.950×10^{-21}	5.261×10^{-21}	5.298×10^{-21}
1	1	0	10.39	10.3170	10.3167	6	1	0	1	980.38	981.80	981.85	1.735×10^{-22}	1.637×10^{-22}	4.385×10^{-22}
1	1	0	10.39	10.3170	10.3167	6	2	0	2	982.51	983.70	983.30	1.460×10^{-21}	2.607×10^{-21}	2.575×10^{-21}
1	1	0	10.39	10.3170	10.3167	4/6	2	2	0	999.41	1000.55	1000.34	3.272×10^{-21}	5.452×10^{-21}	6.531×10^{-21}
1	1	0	10.39	10.3170	10.3167	4/6	2	2	1	1038.92	1040.52	1040.37	1.433×10^{-21}	2.888×10^{-21}	2.333×10^{-21}
2	0	2	3.44	3.4373	3.4374	4/6	1	1	0	986.51	987.65	987.45	3.664×10^{-22}	6.169×10^{-22}	7.514×10^{-22}
2	0	2	3.44	3.4373	3.4374	4/6	1	1	1	1006.27	1007.65	1007.49	1.664×10^{-22}	3.188×10^{-22}	2.450×10^{-22}
2	0	2	3.44	3.4373	3.4374	4/6	2	1	1	1008.56	1009.94	1009.78	8.183×10^{-22}	1.609×10^{-21}	1.233×10^{-21}
2	0	2	3.44	3.4373	3.4374	4/6	2	1	2	988.80	989.94	989.74	1.859×10^{-21}	3.100×10^{-21}	3.706×10^{-21}
2	0	2	3.44	3.4373	3.4374	4/6	3	1	2	992.23	993.36	993.16	1.483×10^{-21}	2.425×10^{-21}	2.833×10^{-21}
2	0	2	3.44	3.4373	3.4374	4/6	3	1	3	1011.98	1013.35	1013.20	6.496×10^{-22}	1.327×10^{-21}	1.022×10^{-21}
2	1	2	12.62	12.5384	12.5385	6	1	0	1	978.15	979.58	979.63	1.650×10^{-22}	1.681×10^{-22}	4.627×10^{-22}
2	1	2	12.62	12.5384	12.5385	4	2	0	2	980.28	981.48	981.07	7.312×10^{-21}	1.304×10^{-20}	1.303×10^{-20}
2	1	2	12.62	12.5384	12.5385	4/6	2	2	0	997.18	998.33	998.12	1.766×10^{-21}	2.978×10^{-21}	3.628×10^{-21}
2	1	2	12.62	12.5384	12.5385	4/6	2	2	1	1036.69	1038.30	1038.15	8.129×10^{-22}	1.550×10^{-21}	1.241×10^{-21}
2	1	2	12.62	12.5384	12.5385	6	3	0	3	983.88	985.29	985.34	1.271×10^{-22}	1.098×10^{-22}	2.762×10^{-22}
2	1	2	12.62	12.5384	12.5385	4/6	3	2	1	1040.13	1041.73	1041.58	1.545×10^{-21}	3.203×10^{-21}	2.591×10^{-21}
2	1	2	12.62	12.5384	12.5385	4/6	3	2	2	1000.61	1001.76	1001.55	3.590×10^{-21}	5.944×10^{-21}	7.062×10^{-21}
2	1	1	12.72	12.6436	12.6432	4	1	0	1	977.89	979.09	978.68	4.379×10^{-21}	7.805×10^{-21}	7.881×10^{-21}
2	1	1	12.72	12.6436	12.6432	6	2	0	2	980.34	981.76	981.81	2.831×10^{-22}	2.657×10^{-22}	7.148×10^{-22}
2	1	1	12.72	12.6436	12.6432	4/6	2	2	0	1036.59	1038.19	1038.05	8.148×10^{-22}	1.554×10^{-21}	1.248×10^{-21}
2	1	1	12.72	12.6436	12.6432	4/6	2	2	1	997.08	998.23	998.01	1.785×10^{-21}	3.011×10^{-21}	3.667×10^{-21}
2	1	1	12.72	12.6436	12.6432	4	3	0	3	983.61	984.80	984.40	2.870×10^{-21}	5.127×10^{-21}	5.025×10^{-21}
2	1	1	12.72	12.6436	12.6432	4/6	3	2	1	1000.51	1001.66	1001.44	3.629×10^{-21}	6.009×10^{-21}	7.140×10^{-21}
2	1	1	12.72	12.6436	12.6432	4/6	3	2	2	1040.03	1041.62	1041.47	1.549×10^{-21}	3.211×10^{-21}	2.598×10^{-21}
2	2	1	40.34	40.0423	40.0417	4/6	2	1	1	951.90	953.33	953.13	2.252×10^{-22}	4.716×10^{-22}	3.933×10^{-22}
2	2	1	40.34	40.0423	40.0417	4/6	2	1	2	971.65	973.32	973.17	5.027×10^{-22}	7.996×10^{-22}	9.130×10^{-22}
2	2	1	40.34	40.0423	40.0417	4/6	3	1	2	975.10	976.77	976.62	9.998×10^{-23}	1.564×10^{-22}	1.748×10^{-22}
2	2	1	40.34	40.0423	40.0417	4/6	3	1	3	955.34	956.77	956.57	4.487×10^{-23}	9.566×10^{-23}	7.959×10^{-23}
2	2	1	40.34	40.0423	40.0417	4/6	3	3	0	1068.03	1069.97	1069.83	7.748×10^{-22}	1.337×10^{-21}	1.108×10^{-21}
2	2	1	40.34	40.0423	40.0417	4/6	3	3	1	1008.73	1010.06	1009.83	1.588×10^{-21}	2.668×10^{-21}	3.262×10^{-21}
rms[c]			0.13	0.0003						1.21	0.22		1.807×10^{-21}	4.905×10^{-22}	

[a]Taken from Ref. 30.
[b]Taken from Ref. 65.
[c]Root mean square deviation (calc-obs).

is not. For the modes ν_4 and ν_6, the mean squared error between the intensities computed by Yachmenev *et al.* ($\text{rms}_{\text{TROVE}} \approx 4.7 \times 10^{-22}$) and the experimental data is much better than for ours ($\text{rms}_{\text{MOLPRO}} \approx 1.8 \times 10^{-21}$). A systematic analysis showed that this effect arises from the vibrational transition dipole moments. Once we use the vibrational transition dipole moments of Yachmenev *et al.*[30] within our calculations, our rovibrational intensities essentially coincide with those of Yachmenev *et al.* Deviations are then in the same

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TABLE VII. Line list for the molecule thioformaldehyde for the ν_4 and ν_6 transitions at 989.36 cm^{-1} and 989.47 cm^{-1} at a temperature of 300 K, resulting in a partition function of Q(300 K) = 6096.4. The point group is C_{2v}, and the nuclear spin statistical weights are 1 for transitions $A_1 \leftrightarrow A_2$ and 3 for transitions $B_1 \leftrightarrow B_2$. Intensities are given in cm^{-1}/(molecule cm^{-2}).

J''	K''_a	K''_c	E''	$E''_{\text{Calc.}}$ [a]	$E''_{\text{Expt.}}$ [b]	ν'	J'	K'_a	K'_c	f	$f_{\text{Calc.}}$ [a]	$f_{\text{Expt.}}$ [b]	I	$I_{\text{Calc.}}$ [a]	$I_{\text{Expt.}}$ [b]
3	0	3	6.88	6.8743	6.8745	4/6	2	1	1	985.36	986.50	986.30	7.152×10^{-22}	1.211×10^{-21}	1.490×10^{-21}
3	0	3	6.88	6.8743	6.8745	4/6	2	1	2	1005.11	1006.49	1006.33	3.301×10^{-22}	6.242×10^{-22}	4.799×10^{-22}
3	0	3	6.88	6.8743	6.8745	4/6	3	1	2	1008.56	1009.94	1009.79	1.123×10^{-21}	2.207×10^{-21}	1.693×10^{-21}
3	0	3	6.88	6.8743	6.8745	4/6	3	1	3	988.80	989.94	989.73	2.574×10^{-21}	4.295×10^{-21}	5.127×10^{-21}
3	1	3	16.01	15.9231	15.9234	6	2	0	2	977.05	978.48	978.53	2.132×10^{-22}	2.263×10^{-22}	6.297×10^{-22}
3	1	3	16.01	15.9231	15.9234	4/6	2	2	0	1033.30	1034.91	1034.77	1.682×10^{-22}	2.952×10^{-22}	2.341×10^{-22}
3	1	3	16.01	15.9231	15.9234	4/6	2	2	1	993.79	994.95	994.73	3.447×10^{-22}	5.926×10^{-22}	7.358×10^{-22}
3	1	3	16.01	15.9231	15.9234	4	3	0	3	980.32	981.52	981.12	1.011×10^{-20}	1.804×10^{-20}	1.803×10^{-20}
3	1	3	16.01	15.9231	15.9234	4/6	3	2	1	997.22	998.38	998.16	3.014×10^{-21}	5.084×10^{-21}	6.196×10^{-21}
3	1	3	16.01	15.9231	15.9234	4/6	3	2	2	1036.74	1038.34	1038.19	1.397×10^{-21}	2.663×10^{-21}	2.130×10^{-21}
3	1	2	16.21	16.1333	16.1329	4	2	0	2	976.69	977.88	977.48	5.738×10^{-21}	1.022×10^{-20}	1.046×10^{-20}
3	1	2	16.21	16.1333	16.1329	4/6	2	2	0	993.59	994.74	994.52	3.523×10^{-22}	6.056×10^{-22}	7.522×10^{-22}
3	1	2	16.21	16.1333	16.1329	4/6	2	2	1	1033.10	1034.70	1034.56	1.689×10^{-22}	2.969×10^{-22}	2.349×10^{-22}
3	1	2	16.21	16.1333	16.1329	6	3	0	3	980.29	981.70	981.75	3.839×10^{-22}	3.576×10^{-22}	9.676×10^{-22}
3	1	2	16.21	16.1333	16.1329	4/6	3	2	1	1036.53	1038.13	1037.98	1.403×10^{-21}	2.678×10^{-21}	2.146×10^{-21}
3	1	2	16.21	16.1333	16.1329	4/6	3	2	2	997.02	998.17	997.95	3.080×10^{-21}	5.196×10^{-21}	6.328×10^{-21}
3	2	2	43.78	43.4793	43.4787	4/6	2	1	1	968.22	969.90	969.74	9.748×10^{-22}	1.571×10^{-21}	1.849×10^{-21}
3	2	2	43.78	43.4793	43.4787	4/6	2	1	2	948.46	949.90	949.70	4.552×10^{-22}	9.344×10^{-22}	7.881×10^{-22}
3	2	2	43.78	43.4793	43.4787	4/6	3	1	2	951.89	953.32	953.12	3.859×10^{-22}	8.085×10^{-22}	6.734×10^{-22}
3	2	2	43.78	43.4793	43.4787	4/6	3	1	3	971.64	973.31	973.15	8.686×10^{-22}	1.383×10^{-21}	1.584×10^{-21}
3	2	2	43.78	43.4793	43.4787	4/6	3	3	0	1005.29	1006.63	1006.39	5.399×10^{-22}	9.275×10^{-22}	1.155×10^{-21}
3	2	2	43.78	43.4793	43.4787	4/6	3	3	1	1064.59	1066.53	1066.39	1.986×10^{-22}	5.069×10^{-22}	4.385×10^{-22}
3	2	1	43.78	43.4798	43.4792	4/6	2	1	1	948.46	949.89	949.69	4.521×10^{-22}	9.288×10^{-22}	7.795×10^{-22}
3	2	1	43.78	43.4798	43.4792	4/6	2	1	2	968.21	969.89	969.73	9.851×10^{-22}	1.591×10^{-21}	1.873×10^{-21}
3	2	1	43.78	43.4798	43.4792	4/6	3	1	2	971.66	973.33	973.18	8.507×10^{-22}	1.349×10^{-21}	1.553×10^{-21}
3	2	1	43.78	43.4798	43.4792	4/6	3	1	3	951.90	953.33	953.13	3.911×10^{-22}	8.182×10^{-22}	6.820×10^{-22}
3	2	1	43.78	43.4798	43.4792	4/6	3	3	0	1064.59	1066.53	1066.39	1.986×10^{-22}	5.069×10^{-22}	4.385×10^{-22}
3	2	1	43.78	43.4798	43.4792	4/6	3	3	1	1005.29	1006.63	1006.39	5.399×10^{-22}	9.275×10^{-22}	1.155×10^{-21}
3	3	1	89.91	89.2000	89.1985	4/6	2	2	0	959.40	961.64	961.49	3.501×10^{-21}	5.594×10^{-21}	6.656×10^{-21}
3	3	1	89.91	89.2000	89.1985	4/6	2	2	1	919.89	921.67	921.46	1.588×10^{-21}	3.306×10^{-21}	2.762×10^{-21}
3	3	1	89.91	89.2000	89.1985	4/6	3	2	1	923.32	925.10	924.89	5.481×10^{-22}	1.159×10^{-21}	9.598×10^{-22}
3	3	1	89.91	89.2000	89.1985	4/6	3	2	2	962.83	965.07	964.92	1.232×10^{-21}	1.942×10^{-21}	2.247×10^{-21}
3	3	0	89.91	89.2000	89.1985	4/6	2	2	0	919.89	921.67	921.46	1.588×10^{-21}	3.306×10^{-21}	2.762×10^{-21}
3	3	0	89.91	89.2000	89.1985	4/6	2	2	1	959.40	961.64	961.49	3.501×10^{-21}	5.594×10^{-21}	6.656×10^{-21}
3	3	0	89.91	89.2000	89.1985	4/6	3	2	1	962.83	965.07	964.92	1.232×10^{-21}	1.942×10^{-21}	2.247×10^{-21}
3	3	0	89.91	89.2000	89.1985	4/6	3	2	2	923.32	925.10	924.89	5.481×10^{-22}	1.159×10^{-21}	9.598×10^{-22}
rms[c]			0.38	0.0008						1.43	0.20		2.000×10^{-21}	4.810×10^{-22}	

[a]Taken from Ref. 30.
[b]Taken from Ref. 65.
[c]Root mean square deviation (calc-obs).

regime (rms ≈ 5.4×10^{-22}). Intensive tests to improve our dipole moment surfaces did not lead to substantial changes, and we were not able to reproduce the values of Yachmenev *et al.* with our own DMS. However, the reason for this lies in the electronic structure calculations and not in the rovibrational calculation. Again, we did not try to increase the quality of the DMS since it would be very demanding for large systems. Table VII compares our vibrational transition moments for the fundamental transitions of H_2CS with those of Yachmenev *et al.* and those of Meyer *et al.*[55] Clearly, the agreement is excellent except for ν_4. As a result, all our intensities with contributions from this transition deviate from the results of Yachmenev *et al.* In comparison to the work of Carter and co-workers,[20] our rovibrational intensities are in better agreement with the experimental data.

While experiments in the gas phase are able to measure frequencies with an accuracy of 10^{-4} cm^{-1} or even better, intensities

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TABLE VIII. Vibrational transition moments for the fundamental modes of thioformaldehyde.

		Transition moment [D]		
State	Symmetry	This work	TROVE[a]	MCTDH[b]
ν_4	B_1	0.1249	0.1652	0.1185
ν_6	B_2	0.0244	0.0235	0.0387
ν_3	A_1	0.0558	0.0566	0.0554
ν_2	A_1	0.0247	0.0242	0.0353
ν_1	A_1	0.0585	0.0593	0.0555
ν_5	B_2	0.0275	0.0283	0.0419

[a]Taken from Ref. 30.
[b]Taken from Ref. 55.

often have a much higher uncertainty; see Ref. 66 and the references therein. Simulations, on the other hand, usually give more accurate intensities, but struggle to achieve frequencies at the precision of the experimental data. Potential energy and dipole moment surfaces have to be computed at a very high level of theory, including various corrections as mentioned above in the context of our DMS. Empirical adjustments of the PES for a better agreement between theoretical and experimental results are common.[66–69] However, for the purpose of this benchmark work, no further effort to improve our line intensities has been undertaken.

C. Rovibrational spectrum of thioformaldehyde

The simulated rovibrational spectrum of thioformaldehyde and its comparison with work from other authors are shown in Fig. 3. The TROVE spectrum and the experimental spectrum have been retrieved from the work of Yachmenev *et al.*[30] Our computed spectrum was restricted to the transitions arising from the ν_4 and ν_6 vibrational modes. The three upper spectra shown in Fig. 3 do not show line broadening. Clearly, the agreement of these three spectra is excellent. Only subtle differences can be seen. For example, in the MOLPRO spectrum, a progression can be seen at about 1080 cm^{-1}, which does not show up in the TROVE and the EXP spectra. We believe that, in the latter two spectra, this progression has been assigned to a different upper vibrational mode and has, thus, been excluded from the spectra by purpose. However, according to our calculations, this progression must be included. Moreover, in the region of about 1040 cm^{-1}, there are two fairly narrow progressions (which, due to the resolution of the spectra, appear just as black Gaussians), which tend to be too strong in the TROVE and MOLPRO spectra in comparison to the EXP spectrum. However, line broadening reduces the intensity of these two progressions significantly. Another subtle difference can be seen in the region between 990 cm^{-1} and 1000 cm^{-1}. While the TROVE and MOLPRO spectra agree quite nicely, the peak at about 1000 cm^{-1} in the EXP spectrum is significantly stronger. The appearance of the spectrum including line broadening effects (bottom) is a completely different one, and the regularity of the progressions is broken in the dense regions of the spectrum, so several transitions become very strong. This is in agreement with the experimental spectrum recorded by Maeda *et al.*,[43] which is not shown here.

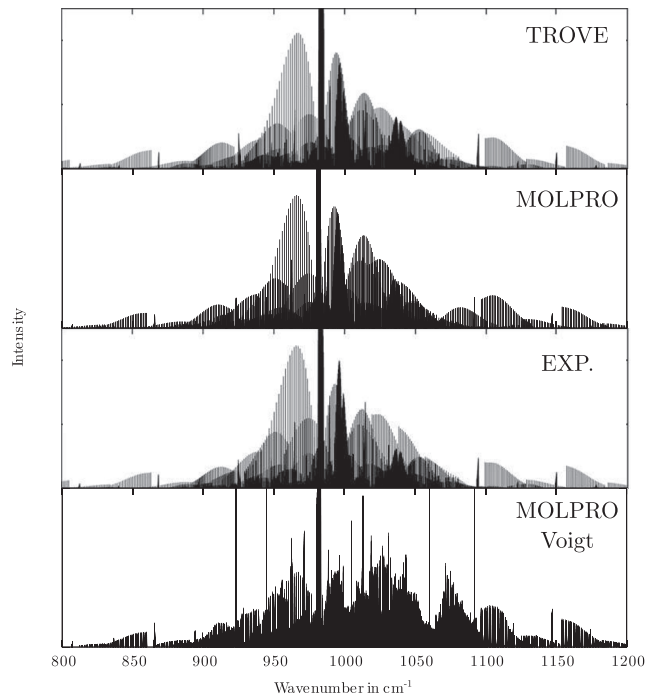

FIG. 3. Part of the rovibrational spectrum of H_2CS for the ν_4 and ν_6 transitions at 989.36 cm^{-1} and 989.47 cm^{-1}, respectively. The TROVE spectrum is taken from Ref. 30 (top), our theoretical spectrum (MOLPRO, second from top), the experimental spectrum reproduced from the work of Yachmenev *et al.*[30] (second from bottom) (all three without line broadening), and our theoretical spectrum (MOLPRO VOIGT) with Voigt line broadening, for T = 300 K and $p = 10^5$ Pa (bottom).

In general, one may not expect a 1:1 agreement for dense regions once line broadening becomes important.

In Fig. 4, we show an analysis of the spectrum with respect to K_a' values. For $K_a' > 7$, the contributions to the spectrum are very small and have, thus, not been shown explicitly. Moreover, the contributions arising from even K_a' values are larger than for odd ones. While the contributions arising from low K_a' values dominate the inner part of the spectrum, the outer regions are mainly controlled by high K_a' values.

The central processing unit (CPU) time requirements for the calculation of this spectrum are dominated by the calculation of the rovibrational intensities. The CPU time for the purely vibrational transitions is negligible, independently, if only fundamental transitions or fundamentals and first overtones and combination bands are considered. The rovibrational energies (up to J = 40) took no longer than 2 min once only fundamentals are considered, while the calculation of the rovibrational intensities of this calculation took about 18 min. The memory requirements were not larger than 4 GB, and thus, these calculations can easily be performed on customary workstations. This shows very clearly that it is computationally feasible to transfer this approach to larger systems. Of course the numbers increase once additional vibrational transitions are included, e.g., the CPU time increases to 9 h and the memory requirements to about 40 GB. All these calculations have been performed on a single core.

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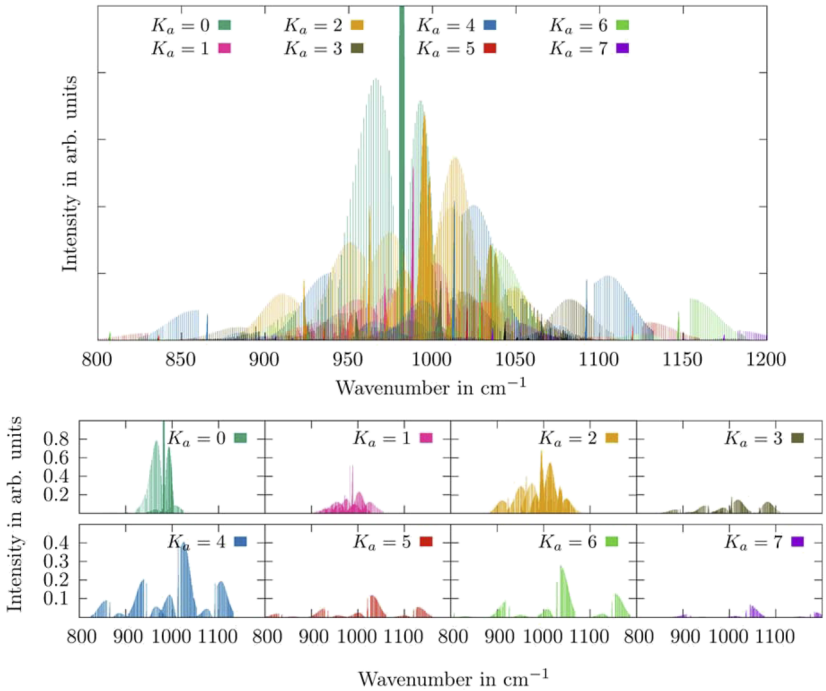

FIG. 4. Part of the rovibrational spectrum of H_2CS using color-coding to distinguish transitions with different upper state quantum numbers $K_a' \leq 7$. The upper graph shows the complete spectrum, as shown in Fig. 3, and in the lower graph, the spectrum is separated by different values of K_a'.

IV. SUMMARY AND CONCLUSIONS

The development of a new program for the automated calculation of rovibrational spectra within the MOLPRO suite of *ab initio* programs has been reported. This program has been designed for the black-box calculation of such spectra in the low energy region up to about 4000 cm^{-1} and temperatures below 500 K. Integral parts of this algorithm are a fully automated construction of the PES and DMS and state-specific configuration-selective VCI calculations, whose eigenvectors are subsequently used in rovibrational calculations and the determination of the rovibrational line intensities. Partition functions can easily be determined from VSCF modal energies without introducing substantial deviations from experimental results. Moreover, Wang combinations and the use of a mode specific rotational basis (MSRB) were found to be important building blocks to assign the rovibrational eigenstates to the correct set of quantum numbers, which reduce the rovibrational correlation matrices in size. While CPU times were found to be moderate for rovibrational state energies, the computational effort increases for the calculation of the corresponding infrared intensities.

The new algorithm has been applied to thioformaldehyde as a benchmark system, which is characterized by strong Coriolis couplings and intensity borrowing. Very good agreement with the available reference data, i.e., line lists and the rovibrational infrared spectrum, has been achieved, proving the capabilities of the new code. Despite the sensitivity of this system, we found it sufficient to restrict the calculations to a rather limited number of vibrational states, which leads to significant computational savings. Line broadening effects, as accounted for by Voigt profiles, were found to have substantial impact on the overall appearance of the spectrum and request highly accurate calculations in order to be comparable with experimental spectra.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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